Graph-Guided Mixture-of-Experts for Unified Multimodal Mental Health Assessment: Affective Constructs Are a Bottleneck, Not a Basis, for Depression

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Abstract

We study whether graph-guided mixture-of-experts (MoE) routing can improve multimodal mental-health assessment when graphs are constructed in a leakage-safe way. We evaluate a unified model on DAIC-WOZ depression detection ($n = 189$ clinical interviews), CMU-MOSEI sentiment and emotion recognition ($n = 22{,}777$), and ChaLearn First Impressions personality prediction ($n = 10{,}000$), combining modality-specific encoders (RoBERTa, WavLM, ViT), gated late fusion, an MMoEEx expert bank, and a KNN-graph GraphSAGE router. Across five leakage-safe graph configurations, a K-sensitivity sweep, and a learned per-task routing weight, evaluated with the same 5-seed replication protocol used throughout this paper, routing does not improve depression detection relative to a non-graph baseline that we independently verify is a real, working classifier (an earlier version was silently capped at chance level by an unused sampling weight; diagnosed and fixed here). A graph-homophily diagnostic localizes why: the routing graph carries no depression-label signal beyond chance while carrying real signal for sentiment, emotion, and personality yet routing still reportably *degrades* personality prediction, showing the failure is not simply signal absence. Pursuing this diagnostic to its representational source yields our central result. Under an identical protocol, we compare three 12-dimensional projections of the same fused representation: a supervised affective-construct projection (the model's own sentiment/emotion/personality heads), a random projection, and PCA. Random projection and PCA are indistinguishable from each other and both exceed the construct projection, which never separates from a label-permutation null; the gap is negative in every one of 5 training seeds and remains negative under a subject-level bootstrap propagating both training and test-sampling uncertainty. Affective-construct projections that are actually achievable at clinical sample sizes whether transferred across corpora or re-derived in-domain degrade depression-relevant variance that an unsupervised projection of the same features retains. A second, independent corpus (MPDD) dissociates why: ground-truth Big-Five personality traits decode depression well above chance, while the same traits, estimated from that corpus's own features by an ordinarily-trained regressor, fail to generalize to held-out subjects at all. The bottleneck is construct *measurement*, not construct *validity*. We additionally explored LLM-based encoders, post-hoc calibration, cross-dataset domain adaptation, and graph-based explanation during development; all of this material is single-seed only, was never extended to this paper's 5-seed protocol, and is reported separately, under an explicit no-claims banner, precisely because a paper whose central finding is that single-seed results can mislead cannot also lean on six single-

seed encoder comparisons as evidence. We release the leakage-safe graph construction protocol, the full multi-seed ablation ladder, and a claim-verification harness that binds every quantitative statement in this paper to a machine-checked artifact.

1 Introduction

Mental health assessment from multimodal signals is a cornerstone challenge in affective computing and clinical informatics. Depression, the primary focus of this work, affects over 280 million people worldwide [1] and is routinely assessed through clinical interviews that combine speech, facial expressions, and verbal content. Prior work tackles this in isolation with task-specific architectures, or enriches it via multitask learning with related affective signals sentiment, emotion, apparent personality [2] under the implicit assumption that supervising a shared representation with these constructs helps rather than hurts the primary clinical task. This paper tests that assumption directly, and finds it does not hold at the sample sizes such studies actually have available.

This paper asks two questions in sequence. First: does graph-guided MoE routing improve multimodal mental-health assessment when graph construction is made leakage-safe and evaluation performed under strict subject-independent splits, replicated across seeds rather than reported from a single training run? We evaluate a unified architecture combining modality-specific encoders, gated late fusion, an MMoEEx expert bank, and a graph-guided GraphSAGE router built on a leakage-safe KNN similarity graph. Across five leakage-safe graph configurations, a K-sensitivity sweep, and a learned per-task routing weight, this routing does not improve depression detection over a non-graph MMoEEx baseline that we independently verify is a real, working classifier (an earlier version of this baseline was silently capped at chance-level DAIC performance by an unused sampling weight in the training data loader; we diagnose and fix this bug as part of this work). A graph-homophily diagnostic comparing real graph edges against random pairing localizes why: the routing graph carries no depression-label signal beyond chance for DAIC, while carrying real signal for MOSEI sentiment/emotion and FI personality; yet routing still reportably degrades FI personality prediction, showing the failure on that task is not simply signal absence.

That diagnostic answers a narrower question than it first appears to. It tells us the routing graph does not encode depression-relevant neighborhood structure in the shared multimodal embedding space, but not why and the natural candidate explanation, that the embedding space is organized around exactly the affective constructs (sentiment, emotion, perceived personality) the model is jointly trained to predict, is directly testable. We pursue it as the paper's second and central question. Under an identical leakage-safe protocol, we compare three 12-dimensional projections of the same fused DAIC representation: a supervised affective-construct projection (the per-task heads' own outputs), an unsupervised random projection, and an unsupervised variance-preserving PCA projection. The random and PCA projections are statistically indistinguishable from each other, and both exceed the construct projection, which never separates from a label-permutation null. The gap is negative in every one of five training seeds, survives dropping near-constant profile dimensions, and remains negative under a subject-level bootstrap that propagates both training-seed and test-subject sampling uncertainty. A separate in-domain re-derivation of the supervision signal, intended to test whether cross-corpus transfer explains the gap, is itself inconclusive under proper seed replication (Section 6) so this specific alternative is not ruled out by that control alone. What the evidence supports is narrower than construct supervision failing in principle: affective-construct projections that are actually achievable at clinical sample sizes, in this setting, degrade depression-relevant variance that an unsupervised projection of the same features retains.

This raises an obvious follow-up: are affective constructs simply uninformative about depression, or is the problem specifically that our estimates of them are unreliable at clinical sample sizes? A second, independent corpus (MPDD) dissociates the two. Ground-truth Big-Five personality scores decode depression well above chance; the same traits, estimated from that corpus's own audio/video features via a regressor trained the ordinary way, fail to generalize to held-out subjects at all (negative held-out R^2 for four of five traits). The construct is informative; our estimate of it is not. We read the DAIC result through this lens: the bottleneck is construct *measurement*, not construct *validity* consistent with, rather than contradicting, the clinical literature linking personality and affect to depression [3, 4].

Figure 1 details the software architecture of our experimental pipeline, moving from data extraction through unimodal baselines into the joint multitask training implementation. To isolate the routing effect, we compare five graph-routing ablation variants (V0–V4) against non-graph MMoEEx and KNN-voting baselines, the latter re-evaluated across 5 seeds after an initial single-seed comparison suggested a DAIC advantage that did not survive proper statistical treatment. We deployed the homophily diagnostic above, together with a K-sensitivity sweep ($K = 5, 10, 15, 20$) and a learned per-task λ weight in place of the fixed default of 0.5; neither recovers a DAIC benefit, consistent with the diagnosis that the routing graph lacks the relevant signal rather than the router failing to exploit signal that is present. We separately explored LLM-based encoders, post-hoc calibration, cross-dataset domain adaptation, and graph-based explanation during development (Section 7); none of this touches the paper's three evidenced claims, all of it is single-seed only, and since this paper's own central finding is that single-seed results in this class of pipeline can mislead we report it separately under an explicit no-claims banner rather than as confirmatory evidence.

This work makes five contributions: (1) a leakage-safe graph construction protocol (three modes inductive, split-local, transductive under strict subject-independent splits, with formal validation against test-label leakage through graph edges); (2) a rigorously investigated, multi-seed negative result for graph-gated MoE routing, with a homophily diagnostic that localizes why rather than reporting an unexplained null; (3) a matched-dimensionality comparison showing that supervising a shared representation with affective constructs destroys depression-relevant variance relative to unsupervised projections of the same features, replicated across five seeds with both training and test-sampling uncertainty accounted for; (4) an independent-corpus dissociation (MPDD) localizing this failure to construct *estimation* rather than construct *validity*; (5) a claim-verification harness binding every quantitative statement in this paper to a machine-checked artifact, together with a documented catalogue of cases, from this project's own history, where that harness changed or retracted a claim a single-seed analysis would have reported with confidence (Section 8).

2 Related Work

Multimodal Fusion. Early fusion for mental health used simple feature concatenation. Later approaches adopted late fusion with learned modality gates or Low-Rank Multimodal Fusion (LMF) [5]. Cross-modal attention mechanisms have also been proposed for depression detection, but in our experimental configuration, we do not observe a benefit from cross-attention over gated fusion (Section 6).

Mixture of Experts. The sparsely-gated MoE layer [6] routes each input to a subset of experts. Multitask MoE (MMoE) extends this with task-specific gates [7], and MMoEEx adds expert exclusivity and orthogonality regularization [8]. We build on MMoEEx but add graph-based routing.

Graph Neural Networks for Routing. GraphSAGE [9] performs inductive representation learning on graphs. We use GraphSAGE as a router: aggregating features from K nearest neighbors to produce a graph context vector that modulates expert weights. Wang et al. [10] similarly combine GNN-based experts with explicit diversity modeling, though as GNN experts rather than a graph-conditioned router over standard MLP experts as here. Whether a KNN routing graph is useful depends on its label homophily/informativeness [11]; when it is not, graph rewiring or learned graph structure [12] are natural remedies we discuss in Section 8. Task-affinity-aware grouping [13] and expert-choice routing [14] are two further candidate fixes we did not yet implement.

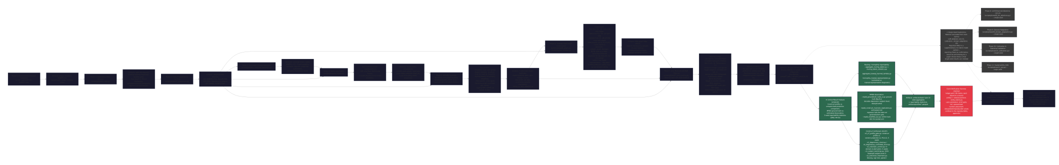

Figure 1: Implementation details of the experimental pipeline. The diagram shows the progression from dataset caching through the non-graph MMoEEx baseline into the Phase 7 multitask epoch loop (5-seed protocol, K-sensitivity sweep, learned routing weight), feeding the central-result analysis (construct-profile comparison, MPDD dissociation, 5-seed reportability statistics) and the claim-verification harness. LLM ablations, domain adaptation, calibration, and XAI (dashed) are single-seed exploratory material, excluded from the paper's claims (Section 7).

LLM-Enhanced Encoders. Large language models have been applied to depression detection via instruction-tuned text encoders and multi-modal audio-language models [15]. We explored LLM-based encoder replacements during development (single-seed, Section 7) but do not report them as a confirmatory result in this paper (Discussion).

Multi-Construct Characterization and Graph-Based Interpretability. Interpretability for mental health assessment has been pursued by jointly modeling depression alongside related affective constructs, and by grounding graph structure in psychologically meaningful units: Vyalla et al. [16] explicitly distinguish personality trait context from acute depression symptoms via a psychologically-grounded, interpretable graph model, related to the affective-construct question this paper investigates directly (Sections 4, 6). For explaining LLM-based reasoning specifically, chain-of-thought and related structured-reasoning techniques have been shown to improve mental-health text classification [17], but can also be unfaithful or reduce reliability when reasoning is inserted into the clinical prediction pipeline itself [18] a caution consistent with our decision to report LLM-based narrative explanation only as single-seed exploratory material (Section 7), not as a confirmed component of the prediction pipeline.

Depression Detection on DAIC-WOZ. The DAIC-WOZ corpus [19] contains 189 clinical interviews with PHQ-8 labels. Prior work spans graph-based text models [20], multimodal behavioral phenotyping [21], and gender-emotion interaction multi-task networks [22]. Reported metrics vary widely by evaluation protocol (F1 vs. AUROC, inclusion of therapist/interviewer prompts, differing label thresholds), so we do not attempt a leaderboard-style numeric comparison and instead report AUROC under our own fixed, leakage-safe protocol throughout.

3 Datasets

DAIC-WOZ contains 189 sessions (107 train, 35 val, 47 test) of human-agent interviews for depression assessment. Each session has a PHQ-8 binary label (depression ≥ 10). We use subject-independent splits: no participant appears in more than one split. Features: RoBERTa-base text (768-dim), WavLM-large audio (1536-dim), eGeMAPS (88-dim), ViT-B/16 video (1536-dim), OpenFace AUs (70-dim).

CMU-MOSEI [23] contains 22,777 utterance-level samples (16,265 train, 1,869 val, 4,643 test). Labels: continuous sentiment and multi-label emotion (happiness, sadness, anger, fear, disgust, surprise). MOSEI is 120$\times$ larger than DAIC, mitigated by temperature-balanced sampling (T=3.0).

ChaLearn First Impressions [24] contains 10,000 video clips (6,000 train, 2,000 val, 2,000 test) with Big-Five apparent personality labels.

Personality is not the same construct as clinical depression; FI serves as auxiliary supervision.

Leakage Protocol. All splits are subject-independent. The graph is constructed separately for each split (split-local mode) or only on training data (inductive mode) to prevent test-label leakage through graph edges.

4 Architecture

Table 1: Notation.

Symbol	Meaning
K	Nearest neighbors (graph)
M	Number of experts (8)
d	Embedding dimension (256)
r	Low-rank bottleneck (8–16)
λ	Graph routing weight
σ_t	Task-specific uncertainty
G	KNN graph
r_i	Graph routing vector

Figure 2 gives an overview of the full pipeline.

Modality Encoders. Each modality $m \in \{\text{text}, \text{audio}, \text{video}\}$ is encoded by a pretrained encoder f_m, producing embedding h_i^m. A projection layer P_m maps to a common $d = 256$-dimensional space:

$$h_i^{m,\text{proj}} = P_m(f_m(x_i^m)) \in \mathbb{R}^{256} \tag{1}$$

Gated Late Fusion. Given projected modality embeddings, we apply a modality mask $m_i \in \{0,1\}^3$ and compute learned gate weights:

$$\alpha_i = \text{softmax}(W_\alpha \cdot \text{concat}(h_i^{\text{text}}, h_i^{\text{audio}}, h_i^{\text{video}}) + b_\alpha) \tag{2}$$

$$h_i^{\text{fused}} = \sum_m \frac{\alpha_i \odot m_i}{\|\alpha_i \odot m_i\|_1 + \epsilon}[m] \cdot h_i^{m,\text{proj}} \tag{3}$$

Each dataset is evaluated under a fixed, dataset-native routing policy chosen from our fusion ablation (Section 6): DAIC uses text-only routing ($h_i^{\text{fused}} = h_i^{\text{text}}$), FI uses video-only routing, and MOSEI uses the full gated fusion above. This is a deliberate architectural choice, not a missing-data fallback: text alone is DAIC's strongest single modality and video alone is FI's, so routing each dataset through its best-performing modality avoids diluting the fused representation with weaker signal on datasets too small to learn a reliable gate over all three. A direct consequence, verified with single-seed post-hoc attribution during development (Section 7), is that a modality outside a dataset's native routing contributes *exactly* zero to that dataset's predictions and to any attribution computed on them (audio and video for DAIC; text and audio for FI) not merely a small or negligible contribution.

MMoEEx Expert Bank. We instantiate $M = 8$ expert MLPs $\{E_1, \dots, E_8\}$ with residual connections, hidden dimension 256 throughout. Two routing configurations are used across our experiments, differing in gating rather than expert size. In the non-graph MMoEEx baseline (Section 6, "MMoEEx vs. Baselines"), experts are hard-isolated per task group: depression uses $\{E_1, E_2\}$, MOSEI sentiment and emotion share $\{E_3, E_4\}$, and personality uses $\{E_5, E_6\}$; E_7, E_8 are allocated but unused under this isolation mask. In the graph-routed variants (V0–V4, our primary reported configuration), we disable hard isolation: task-specific gates $g_t : \mathbb{R}^{256} \to \mathbb{R}^8$ produce a full softmax over all 8 experts, combined with the graph router in log-space (Section 4).

KNN Graph Construction. We construct an undirected KNN graph $G = (V, E)$ over all training samples using cosine similarity. Three construction modes are tested:

- **Inductive (V0, V3)**: Graph on training only. Test nodes connect to K nearest training neighbors.
- **Split-local (V1, V4)**: Separate graphs per split. No cross-split edges.
- **Transductive (V2)**: Graph on all splits. For ablation only.

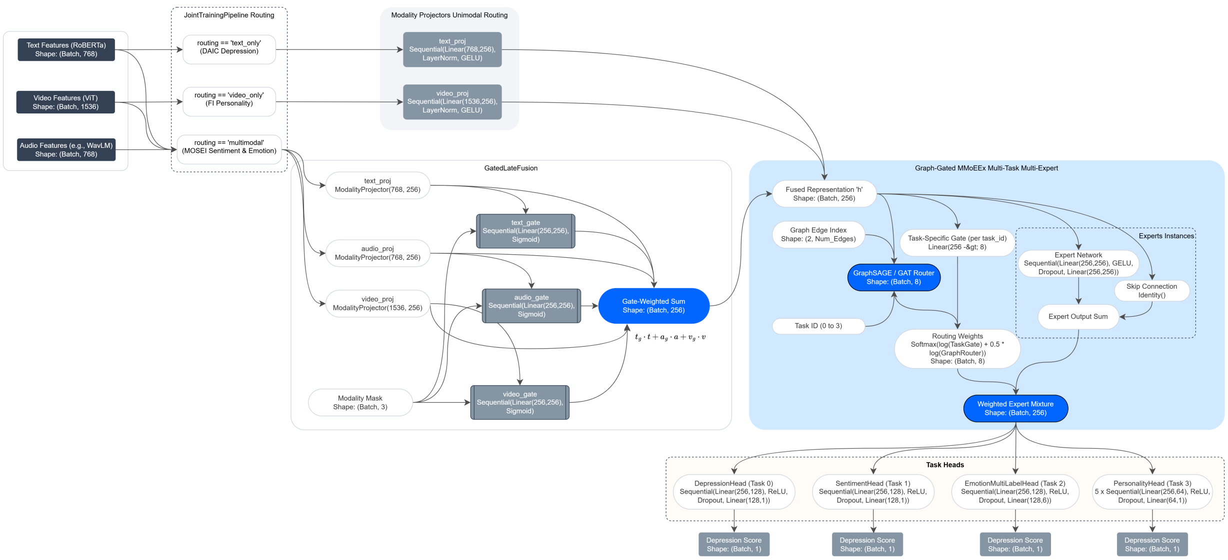

Figure 2: Unified multimodal graph-gated MoE architecture. Modality-specific encoders (RoBERTa/WavLM/ViT) project to a common $d = 256$ space, routed per dataset (text-only for DAIC, video-only for FI, gated late fusion for MOSEI) into h_i^{fused}; the MMoEEx expert bank (8 experts, hidden dimension 256) and a KNN-graph GraphSAGE/GAT router jointly determine expert routing weights in log-space ($\lambda = 0.5$); task heads predict depression, sentiment, emotion, and personality.

The graph in every V0–V4 run (and in the standalone homophily diagnostic, Section 6) is built over a KNN graph on a fresh, randomly-initialized fusion projection of the raw modality features, not the trained model's own fused representation h_i^{fused} used everywhere else in this paper. This is a property of the existing graph-construction code, not something we introduce here, but it was not previously disclosed and materially changes what the homophily diagnostic below can be said to explain; we report both the untrained-embedding version (the one actually used for routing) and a version recomputed on the trained representation (Section 6).

GraphSAGE Router. Two-layer neighborhood aggregation produces a graph routing vector:

$$h_i^{(1)} = \sigma(W^{(1)} \cdot \text{Mean}(\{h_j^{\text{fused}} : j \in \mathcal{N}(i)\} \cup \{h_i^{\text{fused}}\})) \tag{4}$$

$$r_i = \text{softmax}(W_r h_i^{(2)} + b_r) \in \mathbb{R}^M \tag{5}$$

The final expert weights combine task gate and graph router in log-space:

$$\tilde{w}_{t,i} = \text{softmax}(\log g_t(h_i) + \lambda \cdot \log r_i) \tag{6}$$

λ is fixed to 0.5 (not tuned) in all reported graph-routed experiments.

Task Heads. Depression: binary classifier (BCE). Sentiment: regressor (NLL). Emotion: multi-label classifier (BCE). Personality: five regression heads (NLL).

4.1 Construct Profiles and Unsupervised Projections

To test whether affective-construct supervision helps or hurts the shared representation itself, we compare three 12-dimensional projections of the identical fused DAIC representation h_i^{fused} (non-graph baseline, Section 6), under the same downstream fitting protocol (nested-CV logistic regression, subject-independent splits). First, a *construct profile*: the concatenated outputs of the sentiment, emotion, and personality task heads ($1 + 6 + 5 = 12$ dimensions) a supervised, construct-aligned projection. Second, a *random projection*: a fixed random Gaussian matrix applied to the same 256-dimensional fused representation, projected to 12 dimensions an unsupervised control at matched dimensionality. Third, *PCA-12*: the top 12 principal components of the same fused representation, fit on the training split an unsupervised, variance-preserving control that additionally rules out dimensionality reduction per se as the explanation for any effect. Before comparing, we drop per-seed near-constant profile dimensions (train-set standard deviation below a fixed threshold; typically the `happiness` emotion dimension) and regenerate the random projection at the same reduced dimensionality, so the comparison is not confounded by degenerate profile dimensions. All three projections are evaluated identically across 5 independently trained seeds (17, 42, 1337, 2024, 31415); uncertainty is propagated with a subject-level bootstrap (2000 resamples of the DAIC test participants) over the delta between the profile and the matched random projection, averaged across all 5 seeds' fixed predictions within each resample, so that both training-seed and test-subject sampling uncertainty are reflected in the reported interval.

4.2 Independent-Corpus Dissociation (MPDD)

To test whether a DAIC-specific finding reflects construct validity or construct estimation, we use a second, independent corpus, MPDD (Multimodal Depression Detector, Wav2Vec2 and OpenFace features, two age tracks), which uniquely provides *ground-truth* Big-Five personality scores rather than model-estimated ones. We compare two decodings of MPDD depression status: (i) directly from ground-truth Big-Five traits, and (ii) from Big-Five traits *estimated* from MPDD's own raw audio/video features by a ridge regressor trained the ordinary way. Before using estimator (ii)'s predictions in any downstream comparison, we require it to clear a held-out generalization gate (positive test R^2); an estimator that fails this gate is not a valid construct estimate and its downstream predictions are not used (Section 6 reports what happens when this check is skipped). Because MPDD's Young and Elderly tracks are small samples, we additionally use a stratified AUC estimator when pooling them: concordant pairs are counted only *within* a track, removing the contribution of between-track differences to the ranking statistic, which would otherwise inflate or deflate the pooled estimate for reasons unrelated to within-track discrimination.

5 Experimental Setup

Training uses AdamW (lr=3×10^{-4}), cosine annealing, early stopping (patience=20, max 150 epochs). Temperature-balanced sampling (T=3.0) prevents MOSEI dominance. The multitask objective is an unweighted sum of the four task losses; the two regression losses (sentiment, personality) additionally use a homoscedastic-uncertainty NLL form [25] with a single learned σ shared across both:

$$\mathcal{L} = \mathcal{L}_{\text{dep}}^{\text{BCE}} + \mathcal{L}_{\text{emo}}^{\text{BCE}} + \frac{1}{2\sigma^2}\left(\mathcal{L}_{\text{sent}} + \mathcal{L}_{\text{pers}}\right) + \log\sigma \quad (7)$$

Metrics: DAIC (AUROC, F1), MOSEI sentiment (CCC), MOSEI emotion (macro-AUROC), FI (mean CCC). Every result in Sections 6–8 that this paper treats as evidence (the construct-bottleneck comparison, the MPDD dissociation, the graph-routing ablation, the K-sensitivity sweep, the learned per-task λ result, and the homophily diagnostic) is computed across 5 independently trained seeds (17, 42, 1337, 2024, 31415), with reportability gated by a fixed rule (SPEC-H4-03): a delta is described directionally only if it exceeds both the relevant baseline's 95% bootstrap CI half-width and twice the pooled cross-seed standard deviation; otherwise we report it as indistinguishable from seed noise at this sample size rather than assign it a direction. Only the material in Section 7 remains single-seed and is explicitly labeled as such throughout.

Figure 3 shows representative per-task training curves for the V0 configuration: the two large-sample

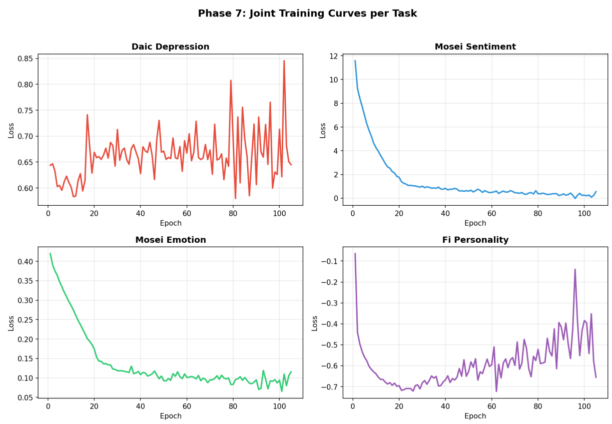

Figure 3: Joint training loss per task (V0 graph model, 105 epochs). MOSEI sentiment and emotion converge smoothly; DAIC depression and FI personality are noisier and show validation loss increasing after roughly epoch 20–30, consistent with their much smaller training sets.

tasks (MOSEI sentiment, MOSEI emotion) converge smoothly, while DAIC depression and FI personality the two small-sample tasks are visibly noisier and begin to overfit after roughly epoch 20–30, motivating the early-stopping choice above. Figure 1 details the implementation of this training procedure end to end, including the non-graph MMoEEx baseline (Phase 5) that precedes it and serves as the comparison point throughout Section 6.

6 Results

Unimodal Baselines. Text is the strongest modality for DAIC and MOSEI (CCC=0.5123); video is strongest for FI (Avg CCC=0.4578). All three datasets beat trivial baselines (Figure 4). The DAIC text-only AUROC we report throughout this paper, including as the reference point for the construct-bottleneck comparison below, is 0.584, a same-protocol recomputation (nested-CV ℓ_2-regularized logistic regression on the raw 768-dim RoBERTa embedding, identical train/test participants used everywhere else in this paper). An earlier project phase reported a different number (0.671) for this same nominal quantity, using its own bespoke script; re-running that script's own unmodified code on the features currently on disk reproduces neither 0.671 nor the recomputed 0.584 (it gives 0.515), and root cause remains undetermined. 0.671 is not used as a baseline, sanity check, or comparison point anywhere in this paper.

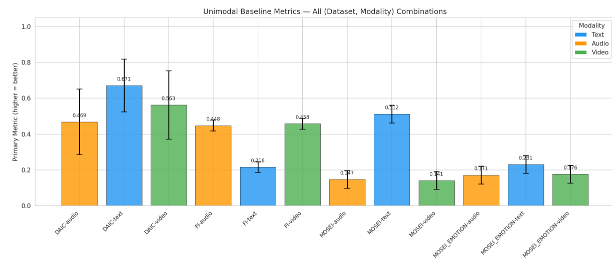

Figure 4: Unimodal baseline metrics with 95% bootstrap CIs across all (dataset, modality) pairs. Overlapping CIs on DAIC reflect the small test set ($n = 47$).

Fusion Ablation. Cross-attention does not outperform gated fusion on any dataset. On DAIC, CrossAttn (AUROC=0.3117) is 0.2229 below text baseline. On MOSEI, Gated (CCC=0.6229) outperforms CrossAttn (CCC=0.5397). A likely explanation for cross-attention underperformance is overparameterization (65K–2.8M parameters vs 57K–827K for gated).

MMoEEx vs Baselines. MMoEEx (hard expert isolation, Section 4) improves FI personality (+0.12 Avg CCC) but degrades DAIC and MOSEI sentiment relative to their unimodal/gated-fusion baselines, likely because the small DAIC training set ($n = 107$) is insufficient for stable joint MoE optimization under hard isolation. Figure 5 shows the resulting routing distribution: each task concentrates weight on a small subset of its assigned experts rather than routing uniformly, indicating the gates do learn task-differentiated behavior even where the aggregate metric declines.

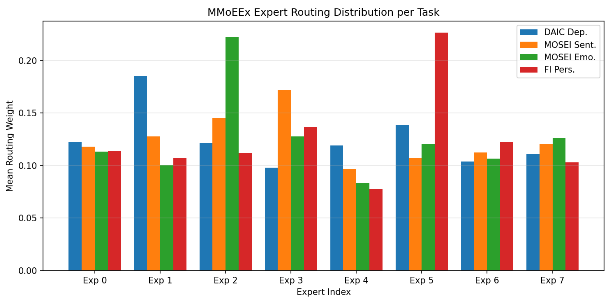

Figure 5: Mean MMoEEx routing weight per expert and task. Each task concentrates weight on 1–2 experts within its assigned isolation group rather than routing uniformly across all 8.

Table 2: Matched-dimensionality comparison at DAIC test AUROC, mean ± std over 5 seeds. Both rows are projections of the identical fused representation, same dimensionality (per-seed, after dropping near-constant profile dimensions), same fitting protocol.

Representation	DAIC AUROC
Random projection (matched dim.)	0.607 ± 0.021
Construct profile (degeneracy-controlled)	0.544 ± 0.018

6.1 Construct-Aligned Projections Underperform Unsupervised Ones

Across 5 independently trained seeds of the non-graph MMoEEx baseline, the 12-dimensional construct profile (Section 4.1) decodes DAIC depression at mean test AUROC 0.524 ± 0.039, never separating from a label-permutation null (per-seed permutation $p \in [0.30, 0.75]$) an absolute claim this design has power only to bound loosely at n_{test}=47. The comparison that is well-powered is relative: a random 12-dimensional projection of the identical fused representation reaches 0.616 ± 0.017, and PCA-12 (unsupervised, variance-preserving, fit on train) reaches 0.610 ± 0.040 indistinguishable from each other, and both above the construct profile in every seed. Table 2 reports the matched-dimensionality comparison after dropping per-seed near-constant profile dimensions and regenerating the random projection at the same reduced dimensionality: the profile reaches 0.544 ± 0.018 against a matched random projection of 0.607 ± 0.021, a delta of -0.063 ± 0.011, negative in all 5 seeds (one-sided sign test p=0.031).

Because all 5 seeds share the same 47 test participants, a cross-seed comparison alone would propagate only training-randomness uncertainty, not test-subject sampling uncertainty. A subject-level bootstrap (2000 resamples of the 47 test participants, delta averaged across all 5 seeds' fixed predictions within each resample) gives mean delta -0.076, 95% CI $[-0.119, -0.037]$, entirely below zero (99.95% of resamples negative) the properly-powered statistic for this claim.

We checked two alternative explanations before treating this as a construct-supervision effect. First, that it is merely dimensionality reduction of any kind: PCA-12 lands within noise of the random projection and clearly above the construct profile the effect is specific to construct alignment, not to projecting to 12 dimensions per se. Second, that it reflects zero-shot cross-corpus transfer (the sentiment/emotion heads are trained on MOSEI, not DAIC): we re-derived sentiment and emotion supervision directly on DAIC transcripts with off-the-shelf classifiers [26, 27] and refit a DAIC-native projector, per seed (Section 4.1). Across all 5 seeds, only 1–3 of 7 re-derived dimensions clear the held-out generalization gate (never a fixed set), itself informative about how reliably affective dimensions are estimable from DAIC's own n=107 training set. Using whichever dimensions pass per seed, the delta between the in-domain profile and a matched-dimensionality random projection is -0.037 ± 0.084 across the 5 seeds 3 of 5 negative, 2 of 5 positive (one-sided sign test p=0.5). This control is **inconclusive, not corroborating**: at this sample size, it does not settle whether domain shift contributes to the main result. The evidence against domain shift as the primary explanation rests on the cross-corpus finding's own 5-seed replication and subject-level bootstrap above, not on this control.

6.2 An Independent Dissociation: MPDD

If affective constructs were simply uninformative about depression, the DAIC result above would be unsurprising. MPDD [28] lets us test this directly, since it provides real (not model-estimated) Big-Five scores. Ground-truth Big-Five decodes MPDD depression at AUROC 0.667 (95% CI $[0.273, 1.000]$, Young track, n=13 test subjects) and 0.925 (95% CI $[0.727, 1.000]$, Elderly, n=13); pooling both tracks with a stratified estimator (Section 4.2) gives 0.857 (95% CI $[0.681, 1.000]$, n=26). These intervals are wide and the ceiling-touching upper bound reflects real imprecision at this sample size; we read this arm as corroborating the clinical literature linking personality to depression [3], not as independently establishing it.

The same construct, *estimated* from MPDD's own raw audio/video features by a ridge regressor trained the ordinary way, fails to generalize at all: held-out R^2 is negative for 4 of 5 traits (worst: Extraversion, R^2=−2.86; best: Agreeableness, R^2=−1.75), despite train-set R^2 of 0.91–0.99 a textbook overfitting signature at this feature-to-sample ratio (1221 features, 184 training subjects). Ground-truth traits are informative; our estimate of them, in this corpus and at this sample size, is not. We take this as the mechanism behind Section 6.1: the bottleneck is construct *measurement*, not construct *validity*.

An early version of this analysis computed depression decoding directly from the overfit MPDD regressor's predictions before checking its held-out R^2, producing a striking-looking result in the opposite direction a large, spurious inversion. We discarded this number once the generalization check surfaced the problem, rather than reporting it as a result (Discussion, Table 5); it motivates a general rule we adopt here: any derived predictor whose output feeds a downstream analysis must clear a held-out generalization check before its output may be consumed the same rule that gates the in-domain DAIC control above.

Graph Routing Ablation. Five graph variants (V0–V4) are compared against the non-graph MMoEEx baseline in Table 3 and Figure 6, mean ± std over the same 5 canonical seeds, with a cell bolded only where the delta from baseline is reportable under the SPEC-H4-03 rule (Section 5); all other deltas are within seed noise at this sample size.

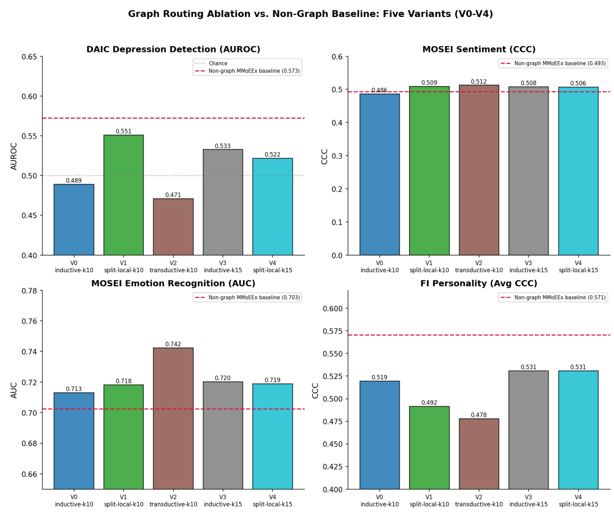

Figure 6: Graph routing ablation vs. the non-graph MMoEEx baseline (red dashed line), all five variants across the 5-seed protocol.

Table 3: Graph routing vs. non-graph MMoEEx baseline, mean ± std over the 5 canonical seeds, real labels and evaluation throughout. **Bold** only where reportable (SPEC-H4-03).

Variant	DAIC AUROC	MOSEI Sent. CCC	FI CCC
Non-graph MMoEEx	*0.541 ± 0.017*	*0.482 ± 0.011*	*0.571 ± 0.006*
V0 (inductive-k10)	0.518 ± 0.027	0.510 ± 0.024	**0.480 ± 0.060**
V1 (split-local-k10)	0.543 ± 0.033	0.498 ± 0.023	**0.481 ± 0.052**
V2 (transductive-k10)	0.595 ± 0.063	0.486 ± 0.019	**0.493 ± 0.050**
V3 (inductive-k15)	0.549 ± 0.057	0.501 ± 0.022	0.507 ± 0.047
V4 (split-local-k15)	0.538 ± 0.051	0.501 ± 0.017	0.506 ± 0.046

Applying the SPEC-H4-03 reportability rule to all 20 (variant × metric) cells on DAIC and MOSEI: **none are reportable** every apparent difference from baseline on DAIC AUROC, MOSEI sentiment CCC, and MOSEI emotion AUROC is within seed noise at n=5 seeds. This supersedes an earlier single-seed version of this table that read "every variant underperforms baseline on DAIC" and "every variant exceeds baseline on MOSEI sentiment" both directional claims that do not survive proper multi-seed statistical treatment (Table 5). FI personality is the exception: 3 of 5 variants (V0, V1, V2) show a reportable *negative* delta ($p < 0.05$, paired t-test), and the remaining two are borderline ($p \approx 0.05$–0.06) graph routing consistently and substantially degrades FI personality prediction, the one directionally robust routing effect in this study. A KNN-voting baseline (label smoothing over the same graph neighborhoods, no learned router), extended to the same 5 seeds, reaches DAIC AUROC 0.623 ± 0.071 numerically above the non-graph baseline, but not reportably so (paired Δ=+0.083, p=0.087); it is, however, reportably *worse* than the non-graph baseline on all three other metrics ($p < 0.02$) simple neighborhood label-smoothing is not a free improvement once measured across seeds, even though a single-seed comparison previously made it look competitive on DAIC alone.

Why does graph routing not help? A homophily diagnostic. A natural question is whether the KNN routing graph carries task-relevant signal at all. The graph actually used to route experts in every V0–V4 run is built over a KNN graph on a separate, randomly-initialized, untrained fusion projection of the raw modality features, not the trained model's own fused representation (Section 4) disclosed here rather than silently carried forward. Table 4 reports both the (5-seed-averaged) untrained-embedding version actually used for routing and a version recomputed on the trained representation.

On the untrained embedding actually used for routing, the 5-seed result is unambiguous and uniform: DAIC shows no signal beyond chance in any of the four leakage-safe variants ($p \in [0.12, 0.78]$), while MOSEI and FI both show strong signal in every variant ($p < 0.0001$). Recomputing the identical diagnostic on the trained representation instead shows a more specific secondary pattern, on the representation not actually used for routing: DAIC signal is graph-topology-dependent (real edges exceed random pairing for the two inductive variants, $p < 0.0001$;

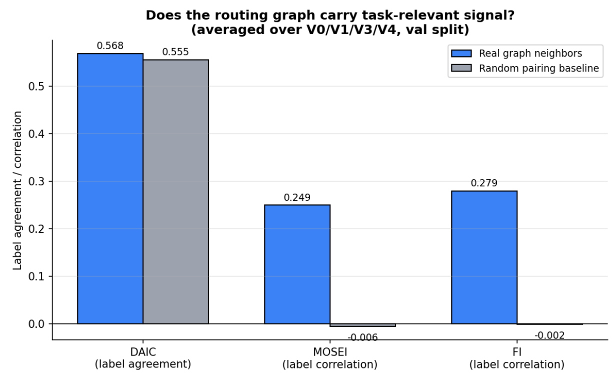

Figure 7: Real graph neighbors vs. random pairing. DAIC shows no signal beyond chance on the untrained embedding actually used for routing; MOSEI and FI both show substantial signal.

split-local variants show no separation, p=0.36 and p=0.94), while MOSEI and FI again show strong signal regardless of topology. Neither version of the diagnostic explains the FI degradation above: FI carries strong real signal by both measures, in every variant, yet routing reportably *hurts* FI prediction rather than merely failing to exploit available signal the router's failure mode on FI is not signal absence.

Two targeted fix attempts. We tried two changes motivated by this diagnostic, both extended to the full 5-seed protocol (reportability per SPEC-H4-03, same non-graph baseline as Table 3); neither recovers a reportable DAIC benefit. First, a $K \in \{5, 10, 15, 20\}$ sensitivity sweep (inductive graph; K=10, K=15 are V0, V3 above) shows no monotonic trend and no configuration's DAIC delta from baseline is reportable (K=5: 0.541 ± 0.048, Δ=+0.001, p=0.97; K=20: 0.526 ± 0.057, Δ=−0.015, p=0.55; Figure 8). K=5 does show a reportable negative FI delta (p=0.04), consistent with the FI-degradation pattern above; K=20's FI delta is borderline (p=0.06).

Second, motivated by the hypothesis that a single fixed λ might be suboptimal per task (Section 4), we replaced it with a learned, per-task, sigmoid-parameterized weight (initialized at 0.5, the fixed de-

Table 4: Real graph neighbors vs. random pairing. "Untrained embedding" is the representation actually used to build every V0–V4 routing graph. "Trained repr." recomputes the identical diagnostic on the corresponding trained checkpoint's own fused representation.

Task (representation)	Real grap	
DAIC, untrained embedding (5-seed mean)	0.564	
DAIC, trained repr. (inductive: V0/V3)	0.619	
DAIC, trained repr. (split-local: V1/V4)	0.555	
MOSEI, untrained embedding (5-seed mean)	0.249	
MOSEI, trained repr. (correlation)	0.485	
FI, untrained embedding (5-seed mean)	0.289	[illegible]
FI, trained repr. (correlation)	0.423	−0.001

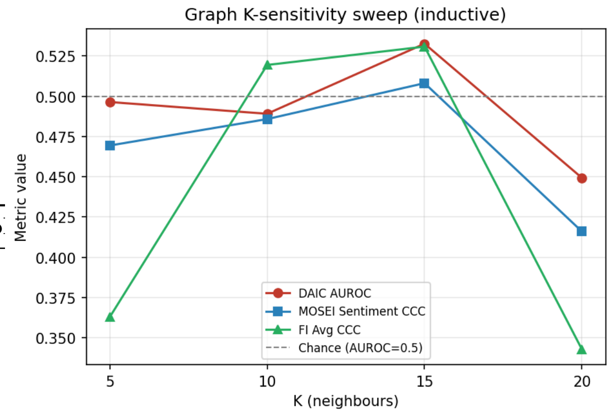

Figure 8: 5-seed mean ± std, inductive graph: no monotonic trend with K, and no configuration's DAIC delta from the non-graph baseline is reportable under SPEC-H4-03.

fault) on the V0 configuration, also extended to 5 seeds. It reaches DAIC AUROC 0.546 ± 0.034 against the fixed-λ V0 baseline's 0.518 ± 0.027 (Figure 9) a delta of $+0.028$ that is *not reportable* under SPEC-H4-03 once cross-seed variance is accounted for (a naive paired t-test alone gives p=0.007, which is exactly why the stricter threshold rule matters). FI CCC moves from 0.480 ± 0.060 to 0.501 ± 0.047, also not reportable. An earlier single-seed comparison (one training run each) read this as producing "a real gain on DAIC" (AUROC 0.489 $\rightarrow$ 0.525); that claim does not survive 5-seed treatment a sixth instance of the reversal pattern in Table 5.

7 Single-Seed Exploratory Material

No claims are made from this section. Everything below was run on a single seed, was never extended to this paper's 5-seed protocol, and touches none of the paper's three evidenced claims (the construct-projection inversion, the MPDD dissociation, or the graph-routing localizability result). We include it because it was part of this project's development history and because a full account of what was tried is more useful to a reader than silence, not because it supports or contradicts anything in Sections 4–6. Reporting it as if it carried the same evidentiary weight as the multi-seed results above would be self-refuting: this paper's central methodological finding is that single-seed results in this class of pipeline can mislead (Section 8, Table 5), and six single-seed LLM-encoder levels are exactly the kind of claim that finding warns against.

LLM-Enhanced Encoders (L0–L5). We replaced classical encoders with LLM-based encoders at each modality: L0 classical baseline (RoBERTa+WavLM+OpenFace); L1 Mistral-7B-Instruct-v0.3 frozen text encoder; L2 L1 with a LoRA adapter; L3 L1+CLAP audio; L4 L1+LLaVA-1.5-7B video; L5 the full LLM stack. On DAIC AUROC, L1–L4 (partial swaps) all underperformed the classical baseline; only L5 (full stack) exceeded it, by $+0.076$. On MOSEI, every level improved over classical on both sentiment CCC (best $+0.140$) and emotion AUROC (best $+0.064$). On FI personality, every level underperformed classical. None of the five Li-vs-L0 DAIC comparisons reached significance by the DeLong test ($p > 0.72$ throughout, n=35) these were reported as suggestive, not confirmed, at the time, before this paper's 5-seed protocol existed to actually test whether a single run's point estimate replicates.

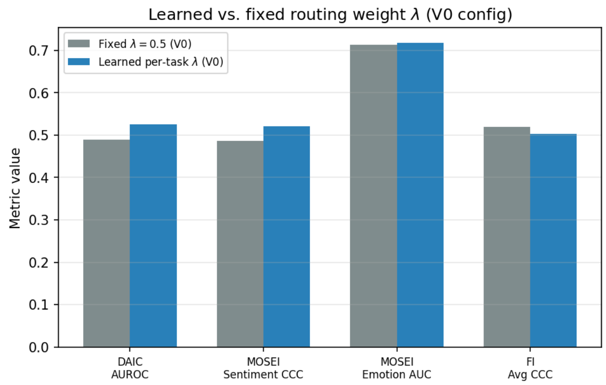

Figure 9: 5-seed mean $\pm$ std: learned per-task λ (V0 config) vs. the fixed default. The DAIC delta is not reportable under SPEC-H4-03 despite a low naive paired-t p-value, illustrating why the stricter cross-seed-variance threshold matters.

Post-hoc Calibration. We applied temperature, Platt, and isotonic scaling to the DAIC classifier (single LLM level). Isotonic regression reduced expected calibration error most (from an uncalibrated range of 0.26–0.35); all three methods are monotonic transforms and leave AUROC-based discrimination unchanged by construction, so this exploration speaks only to probabilistic reliability, not to any claim in this paper.

Cross-Dataset Domain Adaptation. We tested unsupervised domain adaptation (CORAL, MMD, DANN, combined) for transferring FI/MOSEI representations to DAIC depression detection, against a no-adaptation baseline. All four methods underperformed the no-adaptation baseline in 10 of 12 evaluated conditions, single seed. Separately, on the external MPDD benchmark (Wav2Vec2+OpenFace features, distinct from this paper's MPDD ground-truth-vs-estimated dissociation, Section 6.2): a strongly-regularized logistic regression outperformed our architecture on MPDD-Young (AUROC 0.674 vs. 0.545–0.559); zero-shot Young→Elderly transfer collapsed below chance (AUROC 0.395, the Young-trained classifier collapsing to near-constant predictions on Elderly data); and MPDD→DAIC transfer, sharing a Wav2Vec2-family audio encoder, was positive (AUROC 0.551, above a within-DAIC baseline of 0.344) suggestive that encoder-family match matters more than domain similarity for transfer, though untested beyond this single comparison.

Explainability Case Studies. We generated, for 9 case studies (3 per dataset), a multi-construct profile (the same forward pass explained through all four task heads) alongside SHAP per-modality attribution, a GNNExplainer-style subgraph diagram [29], and an LLM-generated (Mistral-7B) natural-language narrative following the GraphXAIN approach [30], used purely as post-hoc explanation of an independently-computed prediction. A faithfulness check held up mechanistically: modalities outside a dataset's native routing (Section 4) showed exactly zero SHAP, perturbation, and counterfactual sensitivity in every one of 90 perturbation tests and 30 counterfactual searches evidence the attribution pipeline reflects the model's actual routing rather than an incidental full-fusion computation, independent of anything about routing quality per se. Rebuilding the explanation graph on the model's own trained representation, hoping to resolve the main homophily diagnostic's DAIC null (Section 6), did not: label agreement on the same 35 DAIC validation samples was 0.463 for the trained representation, *below* both raw concatenated features (0.577) and random pairing (0.537) task-supervised training did not yield a more label-homophilous space here. Chain-of-thought-style reasoning inserted into a clinical prediction pipeline has separately been shown to reduce reliability even as it appears to improve interpretability [18], a caution we take seriously enough to keep these narratives here, rather than presenting them as validated components of the pipeline.

8 Discussion

The construct-bottleneck result and the graph-routing result are related but should not be overstated as the same measurement. The graph actually used to route experts in V0–V4 is built on a separate, untrained projection of raw features, not the trained model's own fused representation that Section 6.1 analyzes directly a property of the existing graph-construction code that we disclose rather than silently carry forward (Section 6). Recomputed on the trained representation instead, the homophily diagnostic shows a more specific pattern than a clean "DAIC absent, MOSEI/FI present" story: DAIC signal is present for inductive graph construction and absent for split-local, while MOSEI and FI both show strong signal regardless of topology. Table 3's 5-seed, reportability-gated results are consistent with a bottleneck account in one place and not another: FI carries strong real signal by both measures in every variant, yet routing reportably degrades FI prediction rather than merely failing to exploit that signal, echoing Section 6.1's finding that a construct-supervised transformation of a representation can destroy or misuse task-relevant structure the representation itself carries; DAIC, by contrast, shows no reportable routing effect in either direction at this sample size, so the two analyses are complementary evidence about representational structure rather than one confirming the other's specific mechanism. Cross-attention fusion underperforms gated fusion on all three datasets (Section 6), likely due to overparameterization (65K–2.8M params vs. 57K–827K for gated) relative to DAIC's 107 training sessions.

8.1 Six Cases Where Methodology Changed the Answer

This paper's central methodological claim is that single-seed, unverified results in this class of pipeline can mislead. Table 5 documents six concrete instances of this, all from this project's own history, each caught by an automated check rather than manual review, rather than an embarrassment to be minimized: every one of these would have been a false or overstated claim had it not been checked, and every check that caught one was cheap relative to the cost of an incorrect published claim.

8.2 Limitations

DAIC's small training set (n=107) and test set (n=47) limit statistical power for absolute claims throughout we distinguish these explicitly from the better-powered paired/subject-bootstrapped comparisons that are this paper's main evidence (Section 6.1). Reliance on frozen pretrained encoders, sensitivity to graph quality when modalities are missing, and the fact that our two routing-specific fix attempts do not exhaust the candidate remedies the homophily diagnostic suggests (a jointly-trained or contrastively-pretrained embedding space for graph construction, graph rewiring/structure learning [12], and task-affinity-aware expert grouping [13] remain untried) are further limitations. The in-domain control (Section 6.1) is inconclusive at its sample size rather than corroborating; the cross-corpus DAIC result and the MPDD dissociation, both replicated across 5 seeds where applicable, carry the paper's main evidentiary weight. We evaluated LLM-based encoders, post-hoc calibration, cross-dataset domain adaptation, and SHAP/graph-based explanation case studies during development (Section 7); none of these touch the inversion, the dissociation, or the localizability result, and given this paper's own repeated finding that single-seed results can mislead (Table 5), we do not report them here as confirmatory and instead include them, explicitly labeled single-seed and exploratory, in Section 7 only. All results are retrospective on public benchmarks and require prospective clinical validation before deployment.

9 Conclusion

We set out to test whether graph-guided mixture-of-experts routing improves unified multimodal mental health assessment. Across five leakage-safe configurations, a K-sweep, and a learned routing weight, it does not improve depression detection, and a homophily diagnostic localizes why: the routing graph carries no depression-label signal beyond chance. Pursuing that diagnostic to its representational source, we find that affective-construct projections (sentiment, emotion, perceived personality) that are actually achievable at clinical sample sizes degrade depression-relevant variance that an unsupervised projection of the same shared representation retains a matched-dimensionality compari-

Table 5: Six cases from this project where a number changed, or was retracted, once checked. Each was caught by an automated mechanism (a generalization gate, a leakage/consistency audit, a multi-seed reportability rule, or a same-protocol recomputation), not by manual re-reading.

Case	Before	After
MPDD personality-decoding inversion	A striking AUROC (predicted profile), computed before checking the regressor's held-out R^2	Discarded: regressor $R^2 < 0$ on held-out data (train R^2 0.91–0.99); the number reflected overfit-model noise, not a real effect (Section 6.2)
In-domain DAIC control	Single checkpoint: negative delta, read as corroborating the cross-corpus inversion	5 seeds: delta -0.037 ± 0.084, 3 of 5 negative, 2 of 5 positive, sign test p=0.5 inconclusive, not corroborating (Section 6.1)
KNN-voting DAIC advantage	Single seed: AUROC 0.630, read as "best DAIC AUROC in the ladder except the non-graph baseline"	5 seeds: +0.083 over baseline, paired p=0.087 not reportable; reportably *worse* than baseline on all three other metrics
Non-graph MMoEEx baseline	DAIC AUROC 0.493, read as a working classifier	Chance-level: a per-sample sampling weight was computed but never passed to the training `DataLoader`; fixed, reaches 0.541 ± 0.017 (5 seeds)
Learned per-task λ	Single seed: DAIC AUROC $0.489 \rightarrow 0.525$, read as "a real gain on DAIC" from replacing the fixed λ=0.5	5 seeds: delta +0.028 vs. fixed-λ V0, not reportable under SPEC-H4-03 (naive paired p=0.007, below the stricter cross-seed-variance threshold)
DAIC text-only AUROC	0.671, used as a sanity anchor for the corrected baseline	Irreproducible: re-running the original script's own unmodified code on current features gives 0.515, not 0.671; same-protocol recomputation gives 0.584 (Section 6, "Unimodal Baselines")

son, replicated across 5 seeds, robust to dropping near-constant profile dimensions, and negative under a subject-level bootstrap propagating both training and test-sampling uncertainty. A companion in-domain re-derivation, designed to test whether cross-corpus transfer explains this gap, is itself inconclusive under proper seed replication (3 of 5 seeds negative, 2 of 5 positive; Section 6.1) so the claim here is about achievable construct estimation in this regime, not about construct supervision failing in principle. An independent corpus (MPDD) localizes a convergent failure specifically to construct *estimation*, for a different reason (data starvation rather than cross-corpus transfer): ground-truth Big-Five traits decode depression consistent with the clinical literature, while traits estimated from the same corpus's own features do not generalize at all. That two genuinely different mechanisms cross-corpus transfer on DAIC, data starvation on MPDD both degrade construct estimation is what carries the general claim. The bottleneck is construct measurement, not construct validity a position that agrees with the psychological literature on affect and depression while explaining why a machine-learning pipeline built on that literature can still fail. We recommend, as a general practice for this class of pipeline: validate any intermediate predictor on held-out data before consuming its output downstream, prefer variance-preserving (unsupervised or randomly-projected) bottlenecks over construct-supervised ones at small clinical sample sizes, and report an unsupervised-projection control alongside any supervised-representation claim. We release the leakage-safe graph construction protocol, the full multi-seed ablation ladder, and a claim-verification harness that binds every quantitative statement in this paper to a machine-checked artifact.

Ethics Statement. This system is intended as a decision-support tool only and should not replace clinical diagnosis. All results require prospective validation before clinical deployment. Apparent personality and clinical depression are distinct constructs and should not be conflated.

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